

Atlantic Sea Scallop Commercial Catch (TOR 2)

*A working paper for the 2025 Atlantic sea scallop Research Track stock assessment
April 7, 2025*

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TOR 2: Estimate catch from all sources including landings and discards. Describe the spatial and temporal distribution of landings, discards, and fishing effort. Characterize the uncertainty in these sources of data.

1. Overview

The sea scallop fishery in the US Exclusive Economic Zone (EEZ) is managed under the Atlantic Sea Scallop Fishery Management Plan (FMP), first implemented on May 15, 1982. An overview of important management measures is given in Table 2.1. A detailed account of scallop fishery management actions and applications of science into management advice is contained in the Working Paper for TOR2.

From 1982 to 1993, the primary management control was a minimum average meat weight requirement for landings, commonly known as the “meat count” requirement. In 1984, Georges Bank was divided into US and Canadian EEZs (Figure 2.1); prior to this time, US and Canadian vessels fished on both sides of the current boundary.

Amendment 4 of the Sea Scallop Plan (NEFMC 1993), implemented in 1994, changed the management strategy from meat count regulation to limited access, effort control and gear regulations for the entire US EEZ. Limited access permits were issued to vessels with a history in the fishery; no new permits have been issued since. Restrictions were gradually made on days-at-sea (DAS), minimum ring size, and crew limits. DAS have been reduced from over 200 in fishing year 1994 to 120 during 1999-2003 to 24 in open areas in 2018-2022, and finally to 20 in 2023. Note that the scallop “fishing year” used by management was historically March-February, but this was changed to April-March starting in April 2018; however, unless otherwise stated, this assessment uses calendar years.

The minimum size of the rings in the dredge bag was gradually increased from 76 mm (3”) in 1994 to 83 mm (3.25”) in 1995, 89 mm (3.5”) during 1996-2004 and 102 mm (4”) since December 2004. The minimum size of the twine top mesh has also been increased from 15 to 25 cm (6” to 10”) since December 2004; while this measure is mainly to reduce bycatch of finfish, it also likely allows some small swimming scallops to escape. Finally, the crew size on a vessel has been restricted, usually to seven persons, in order to incentivize fishing for larger scallops (since most scallops must be shucked at sea by regulation, a limited crew can process a greater weight of large scallops than smaller ones per unit time). In addition to these measures, three large areas on Georges Bank and Nantucket Shoals were closed to groundfish and scallop

fishing in December 1994 (Link et al. 2005; Figure 2.1). Scallop biomass increased about 20-fold in these areas between 1994-2004 (Hart and Rago 2006). Two areas in the Mid-Atlantic were closed to scallop fishing in April 1998 for three years in order to similarly increase scallop biomass and mean weight.

Sea scallops were formally declared overfished in 1997, and Amendment 7 was implemented during 1998 with more stringent DAS limitations and a mortality schedule intended to rebuild the stocks within ten years. Subsequent analyses conducted in 1999, using an early version of the Scallop Area Management Simulator (SAMS) model to take into account the rebuilding effects of closed areas, indicated that the stocks would rebuild with less severe effort reductions than called for in Amendment 7, so the DAS schedule was thus modified. A combination of closures, effort reduction, gear and crew restrictions led to a rapid increase in biomass (Murawski et al. 2000, Hart and Rago 2006, Hart et al. 2013), and sea scallops were declared to be rebuilt in 2001.

Prior to 2004, there were a number of *ad hoc* area management measures, including the Georges Bank and Mid-Atlantic closures in 1994 and 1998, limited reopenings of portions of the Georges Bank areas between June 1999 and January 2001, and reopening of the first Mid-Atlantic rotational areas in 2001. A new set of regulations was implemented as Amendment 10 (NEFMC 2003) during 2004. This amendment formalized an area based management system (Hart 2003, Hart and Rago 2006, O’Keefe and NEFMC PDT 2022), with provisions and criteria for new rotational closures, and separate allocations (in DAS or total allowable catch [TAC]) for reopened closed areas and open areas (also referred to as ‘open bottom’). The three Georges Bank closed areas were divided into access areas, where fishing is periodically permitted, and long-term closures, where no scallop fishing is permitted. Some of the long-term closures were reopened to scallop fishing in 2018.

Unlike the Georges Bank closures, which applied to all scallop and groundfish fishing, the Mid-Atlantic rotational areas are specific to the scallop fishery. Two areas (Hudson Canyon South and Virginia Beach) were closed in 1998 and then reopened in 2001. Although the small Virginia Beach closure in the far south of the scallops range was unsuccessful, scallop biomass built up in Hudson Canyon access area while it was closed, and substantial landings were obtained from Hudson Canyon during 2001-2007. This area was again closed in 2008, reopened in 2011 and closed for a third time in 2014, and reopened in 2015. A third rotational closure, the Elephant Trunk access area east of Delaware Bay, was closed in 2004 after extremely high densities of small scallops were observed in surveys during 2002 and 2003. About 30,000 mt of scallops worth about \$500 million were landed from that area after it was reopened in 2007. It was closed again in December 2012 after high numbers of small scallops were again observed in surveys. A portion of this area was reopened in 2015, and the remainder was reopened in 2017. A fourth access area, Delmarva, directly south of the Elephant Trunk area, was initially closed in 2007, reopened in 2009, closed in 2012 and reopened in 2014. In 2016, the Delmarva, Elephant Trunk, and Hudson Canyon areas (Figure 2.1) were combined into the Mid-Atlantic access area (MAAA), which remained open to scallop fishing in subsequent years. However, a decrease in scallop productivity in the region led managers to revert the Delmarva portion of the MAAA to open bottom in 2018, and revert the entire remaining MAAA back to open bottom in 2022. In the same year, the new New York Bight rotational area was implemented and closed to protect growth of scallop in the area.

Most landings have come from about 350 vessels with limited access permits. Two types of allocation are given to each of these vessels. The first are trips (with a trip limit, typically 12,000-18,000 lbs or 5443-8165 kg meats) to specified rotational access areas that had been closed to scallop fishing in the past. The second are DAS, which can be used in areas outside the closed and access areas (Table 2.1). Vessels fishing under DAS are restricted to a 7-man crew and must shuck their scallops at sea in order to limit their processing power.

The remainder of landings come from vessels operating under “General Category” permits that are restricted to a certain amount per trip (272 kg meats or 600 lbs per trip until 2021, increased to 363 kg meats 800 lbs per trip in fishing year 2022), with a maximum of one trip per day. Landings from these vessels were less than 1% of total landings in the late 1990s, but increased to about 10% of landings during 2007-2009, and currently constitute about 6% of total landings. This type of permit had been open access, but was converted to an individual transferable quota (ITQ) fishery in March 2010.

The primary port in the sea scallop fishery is New Bedford, MA, which accounts for over 75% of landings in 2022 and 2023. With scallop distribution and fishing effort shifting north over time, other principal ports have changed in recent years, with Point Judith, RI and Gloucester, MA, becoming more important at the expense of Cape May, NJ, and Seaford, VA. Lesser amounts of scallops are landed in many ports from North Carolina to Maine. Toothless offshore (New Bedford style) scallop dredges are the main gear type in all regions, although otter trawls are used to some extent in the Mid-Atlantic, and a small fraction of the catch in the Gulf of Maine comes from divers. A typical limited access vessel tows two 4-4.6 m dredges, but some limited access vessels are restricted to a single 3.2 m dredge (termed a “small dredge permit”). Most general category vessels also use a single 3.2 m or smaller dredge, but some use otter trawls in the Mid-Atlantic. Most bycatch of sea scallops in other fisheries occurs in otter trawls, where the target species are squid, flounders, and other groundfish, although this is relatively small compared to the directed fishery. Recreational catch is negligible.

2. Landings

Prior to 1994, landings and effort data were collected during port interviews by port agents and based on dealer data. Since 1994, commercial data have been available as dealer reports (DR) and in vessel trip report (VTR) logbooks. DR give landings, but not area fished, and have reported landings by market category since 1998. VTR data contain information about area fished, fishing effort, and retained catches of sea scallops. A standardized area allocation (AA) method (Wigley et al. 2008) for matching DR to VTRs and assigning areas to landings was used to allocate landings to region for 1994-2019. In 2020, the Catch Accounting and Monitoring System (CAMS; O’Keefe et al. 2023) became the single comprehensive data source for all northeast commercial catch (landings and discards), consolidating information from several databases (including DR, VTR) and the AA process.

US landings data were stratified by region into four groups: Mid-Atlantic (statistical areas ≥ 600 and 539), Southern New England (statistical areas 533-534, and 536-538), Georges Bank (statistical areas 520-526, 541-543, 561-562, and 551-552), and the Gulf of Maine

(statistical areas 463-464, 467, 500, and 511-515). Landings from unknown areas were prorated to each region. Note that landings from Block Island (statistical area 539) are now included with Mid-Atlantic to be consistent with how surveys are considered in the assessment; this differs from previous assessments where landings from Block Island were included with Southern New England. Data were also stratified by gear into three groups: trawls (otter trawls and beam trawls), dredge (scallop dredge, rakes, clam/quahog dredge, other dredge), and other (all other gears). All 1964-2023 landings data were updated to incorporate any data corrections (data retrieval date 07/17/2024); 1887-1963 landings are from the 2018 stock assessment (NEFSC 2018).

Most landings of sea scallops only retain the adductor muscle, or “meat”, although there is a small market for roe-on scallops. If not otherwise specified, landings in this assessment will be in terms of meat weight.

Sea scallop landings in the US increased substantially after the mid-1940s (Figure 2.2), with peaks occurring around 1960, 1978, 1990, 2004, and 2020. Maximum US landings were 29,109 mt meats in 2004. US landings during 2001 – 2012 were all over 20,000 mt and landings in 2019 were 27,649 mt; the maximum in the 20th century was 17,246 mt in 1990. Landings in 2023 were 12,699 mt meats.

Landings from the Georges Bank and Mid-Atlantic regions have dominated the fishery since 1964 (Table 2.2; Figure 2.3). US Georges Bank landings had peaks during the early 1960s, around 1980 and 1990, but declined precipitously during 1993 and remained low through 1998. Landings from Georges Bank during 1999-2004 were fairly steady, averaging almost 5,000 mt annually, and then increased in 2005-2006, primarily due to reopening of portions of the groundfish closed areas to scallop fishing. Georges Bank landings again increased in 2012-2013, this time mainly due to shift of open effort from the Mid-Atlantic to Georges Bank, to take advantage of large year classes in the latter region but declined in the next few years. Georges Bank landings peaked again in 2018 and 2019 at over 15,000 mt, the largest values in the time series from this region, driven by the exceptionally large 2012 and 2013 year classes. In most recent years, landings from this region have remained high (above 10,000 mt).

Prior to the mid-1980s, Mid-Atlantic landings were generally lower than those on Georges Bank. Mid-Atlantic landings during 1962-1982 averaged less than 1,800 mt per year. An upward trend in both recruitment and landings became evident in the Mid-Atlantic in the mid-1980s and continued into the early 2000s. Landings in this region peaked in 2004 at 23,565 mt before gradually declining since, reflecting the poor 2007-2009 year classes, below average recruitment in the Mid-Atlantic since 2013, and concomitant effort shift onto Georges Bank. Landings from the Mid-Atlantic in 2023 were under 700 mt, reminiscent of levels in the early 1970s.

Landings from other areas (Gulf of Maine and Southern New England) were minor in comparison. Gulf of Maine landings represented less than 3% of the total US sea scallop landings in most recent years but have increased slightly to over 7% of total US landings in 2022 and 2023. Maximum landings in the Gulf of Maine were 1,616 mt during 1980 but trended downward afterwards through 2009, when landings were just 71 mt. Landings from the Gulf of Maine have been increasing in the most recent years; landings in 2022 were the highest

since 1981 at 1,096 mt.

Dealer data (landings) have been reported by market categories (under 10 meats per pound, 10-20 meats per pound, 20-30 meats per pound etc) since 1998 (Figure 2.4). These data indicate a trend towards larger sea scallops in landings. While nearly half the landings in 1998 were in the smaller market categories (more than 30 meats per pound), 75% or more of recent landings were below 20 count and about 99% were below 30 count. In the late 1990s, Mid-Atlantic and Georges Bank scallop landings consisted of a rather uniform distribution of different market sizes. Starting in the early 2000s, the 20-30 count (15-23 g) market category dominated landings. Around 2005, the 10-20 count (23-45 g) market category became dominant indicating the presence of older larger scallops in the catch, a result of rotational management implementation. The largest scallops (U10, >45 g) also became more prevalent during that time, starting in the Mid-Atlantic and shifting to Georges Bank as the spatial distribution of the stock and effort changed. Market categories in the Gulf of Maine do not show this pattern as strongly, which is expected given the lack of substantial rotational areas in this region. There is also an unclassified market category reported, which likely refers to scallops landed in shell.

3. Fishing effort and LPUE

Fishing effort (measured as days absent) (Figure 2.5) shows the increasing importance of the Mid-Atlantic region starting in the mid-1970s, peaking in the early 2000s and decreasing on average since then. In contrast, the importance of Georges Bank has increased as effort shifted to that region. The Gulf of Maine appears to have had significant effort prior to 2000 but this is primarily due to the presence of vessels targeting sea urchins that also land sea scallops. Overall, effort from the Gulf of Maine and Southern New England regions is minimal.

Landings per unit effort (LPUE) (Figure 2.6) were computed as landings per day fished, where days fished represent the time in days that the gear is fishing. This was obtained from the port interview records from larger vessels prior to 1994 and from at-sea observers on limited access vessels since 1994. LPUE showed a general downward trend from the beginning of the time series to around 1998, with occasional spikes upward due to strong recruitment events. LPUE increased considerably since then as the stock recovered. Note the close correspondence in most years between the LPUE in the Mid-Atlantic and Georges Bank, probably reflecting the mobility of the fleet; if one area has higher catch rates, it is fished harder until the rates are equalized. Although comparisons of LPUE before and after the change in data collection procedures during 1994 need to be made cautiously, there is no clear break in the LPUE trend in 1994.

Fishing effort (days fished) was computed as the quotient of Landings/LPUE. Effort is thus in units of days fished on limited access vessels; general category vessels, which usually only fish with one small dredge, would likely fish for several days to account for a single “day fished” under this metric. Effort in the US sea scallop fishery generally increased from the mid-1970s to about 1991, and then decreased during the 1990s, first because of low catch rates, and later as a result of effort reduction measures (Figure 2.7). Effort increased in the Mid-Atlantic during 2000-2007, initially due to reactivation of latent effort among limited access vessels, and then

due to increases in general category effort. Total effort since 2007 has been variable and decreasing slightly, with some shifts between regions.

Data from vessel monitoring systems (VMS) that are required on all scallop vessels fishing on Georges Bank and the Mid-Atlantic can give detailed spatial information on fishing activity. The VMS gives the positions of the vessels every half hour, from which average speed can be calculated. These data were then filtered to eliminate times where the vessels were not fishing (either steaming or simply shucking scallops without gear in the water, Palmer and Wigley 2009), and the resulting spatial distributions were plotted (Figure 2.8).

4. Discards and discard mortality

Sea scallops are sometimes discarded on directed scallop trips because they are too small to be economically profitable to shuck, or because of high-grading, particularly during access area trips. Ratios of discard to total catch (by weight) have been recorded by sea samplers aboard commercial vessels since 1992, though sampling intensity on non-access area trips was low until 2003.

Discarded sea scallops may suffer mortality on deck due to crushing, high temperatures, or desiccation. There may also be mortality after they are thrown back into the water from physiological stress and shock, or from increased predation due to shock and inability to swim or shell damage (Veale et al. 2000, Jenkins and Brand 2001). Murawski and Serchuk (1989) estimated that about 90% of tagged scallops were still living several days after being tagged and placed back in the water. Total discard mortality (including mortality on deck) is uncertain but has been estimated as 20% in previous assessments (e.g., NEFSC 2010, NEFSC 2018). However, discard mortality may be higher in the Mid-Atlantic during the summer due to high water and deck temperatures, and likely strongly depends in both regions on crew practices; scallops returned to the water promptly have much higher chances of survival than ones left on deck for longer periods.

Discard estimates for 1989-2023 are presented in Table 2.3. Discards for 1989-2017 were calculated using the Standardized Bycatch Reporting Methodology (SBRM; Wigley et al. 2007). This approach uses a combined d/k_{all} ratio estimator (Cochran 1963), where d is discarded pounds of sea scallops and k_{all} is kept pounds of all species, calculated from the Northeast Fisheries Observer Program (NEFOP) data. Discard weight was derived by multiplying the d/k_{all} ratio of each fleet by the corresponding dealer or VTR landings. Additional details regarding the discard estimates are available in previous assessment reports for 1989-2009 (Appendix B2 in NEFSC 2014) and 2010-2017 (Appendix A4 in NEFSC 2018). Discards for 2018-2023 are from CAMS, which provide improved estimates. The original (SBRM) approach separated observer information from catch, and then matched discard rates to unobserved trips based on stratification, such that different gear table definitions sometimes led to situations where observed strata did not exist in the catch or vice versa. In contrast, CAMS joins the observer and catch information prior to estimation, which leads to improved discard estimates.

On average, discards were 5.2% of total landings for 1989-2023 and 5.9 % for 2010-2023 (Table 2.3). By region, most discarding occurred in Southern New England (12.9% average for 2010-2023), especially during 2014-2020 when discards were close to 20% of landings, ranging up to 27.2 % in 2020. Discarding in the Mid-Atlantic varied over time with low periods representing less than about 5% of landings, and high periods (1994, 2001-2004, 2016, 2023) representing about 10-15 % of landings. Discarding in Georges Bank was low (<5%) or moderate (5-10%) in most years, with a few years (e.g., 2010, 2020) when discarding was >10%, with a peak of 20% in 2000. The smallest proportion of discarding occurred in the Gulf of Maine region where discards were typically less than 4% until 2021, with a little higher in 2022 and 2023 (4.9% and 5.5%, respectively).

5. Incidental mortality

Scallop dredges likely kill and injure some scallops that are contacted but not caught, primarily due to damage (e.g., crushing) caused to the shells by the dredge. Caddy (1973) estimated that 15-20% of the scallops remaining in the track of a dredge were killed. Murawski and Serchuk (1989) estimated that less than 5% of the scallops remaining in the track of a dredge suffered non-landed mortality. Caddy's study was done in a relatively hard bottom area in Canada, while the Murawski and Serchuk study was in sandy bottom off the coast of New Jersey. It is possible that the difference in indirect mortality estimated in these two studies was due to different bottom types (Murawski and Serchuk 1989). A 2017 study estimated somewhat lower incidental mortality rates of 2.5% in the Mid-Atlantic and 8% on Georges Bank (Ferraro et al. 2017). Two other unpublished studies presented during working group meetings suggest similar rates (Bochenek et al., Smolowitz et al.).

In order to use the above estimates to relate landed and non-landed fishing mortality in stock assessment calculations, it is necessary to know the efficiency e of the dredge (the probability that a fully recruited scallop in the path of a dredge is captured). Similarly, the fraction of scallops that suffer mortality among sea scallops in the path of the dredge but not caught is denoted by c . The ratio R of scallops in the path of the dredge that were caught, to those killed but not caught is:

$$R = \frac{e}{c(1-e)} \quad (1)$$

If scallops suffer direct (i.e., landed) fishing mortality at rate F_L , then the rate of indirect (non-landed) fishing mortality will be (Hart 2003):

$$F_I = F_L / R = \frac{1}{e} F_L c (1 - e) \quad (2)$$

Assuming $c = 0.025$ and $e = 0.6$ for the Mid-Atlantic and $c = 0.08$ and $e = 0.5$ for Georges Bank gives estimates of F_I of about 0.02 for the Mid-Atlantic and 0.08 for Georges Bank. Using an estimate of $c = 0.04$ from Murawski and Serchuk (1989) for the Mid-Atlantic and $c = 0.12$ from Caddy (1973) for Georges Bank gives estimates of incidental mortality of 0.03 and 0.12, respectively. The CASA model does not explicitly model discarding or discard mortality of small

scallops, although it is included in the SYM and SAMS model. Because of this, the working group agreed to set $c = 0.05$ in the Mid-Atlantic and $c = 0.1$ on Georges Bank in the SYM and SAMS models, and $c = 0.06$ and 0.11 , respectively, in the CASA model.

The above calculations are based on the assumption that the scallop is fully selected to the gear. If that is not the case, equation 2 becomes:

$$F_I(h) = \frac{1}{e} F_L c (1 - e^{-q(h)}) \quad (3)$$

where $q(h)$ is the catchability of commercial gear on a scallop of shell height h . We took $q(h)$ to be of the form:

$$q(h) = q_0 s(h)$$

where q_0 is taken as 0.5 on Georges Bank and 0.6 in the Mid-Atlantic, and $s(h)$ is commercial selectivity as estimated by the CASA model.

6. Commercial shell height data

Since most sea scallops are shucked at sea, it has often been difficult to obtain reliable commercial size compositions. Port samples of shells brought in by scallopers have been collected, but there are questions about whether the samples were representative of the landings and catch. Port samples taken during the meat count era often appear to be selected for their size rather than being randomly sampled, and the size composition of port samples from 1992-1994 differed considerably from those collected by at-sea observers during this same period. For this reason, commercial size compositions from port samples after 1984 when meat count regulations were in force are not used in this assessment.

Sea samplers (observers) have collected shell heights of kept scallops from commercial vessels since 1992, and discarded scallops since 1994. Although these data are likely more reliable than that from port sampling, they still must be interpreted cautiously for years prior to 2003 (except for the access area fisheries) due to limited observer coverage. Except for 2006, observer coverage rates have been over 5% since 2003, and have been over 10% in some years.

Shell heights from port and sea sampling data indicate that sea scallops between 70-90 mm often made up a considerable portion of the landings during 1975-1998, but sizes selected by the fishery have increased since then, so that scallops less than 90 mm were rarely taken since 2002 (Figure 2.9).

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Tables

Table 2.1. Key allocations and management measures for the sea scallop fishery, including open area days at sea (DAS) allocations for full time limited access vessels and access area allocations (in kg meats) to Georges Bank (GB) and Mid-Atlantic (MA) access areas (Acc). Prior to Amendment 10 (2004), access area allocations were optimally obtained in exchange for a certain number of DAS. Abbreviations in this table include individual fishing quota (IFQ) and annual catch limit (ACL).

Year	DAS	GB Acc	MA Acc	Other key measures
1982-93	NA	NA	NA	Open access, meat count regulation
1994	204	0	0	Limited access, GB closures
1995	182	0	0	83 mm rings
1996	182	0	0	89 mm rings
1997	164	0	0	Overfishing/overfished declaration
1998	142	0	0	First MA rotational closures
1999	120	13,608	0	
2000	120	27,216	0	
2001	120	0	23,133	Fishery declared rebuilt
2002	120	0	24,494	
2003	120	0	28,577	
2004	42	8,165	32,659	102 mm rings, rotational management
2005	40	16,329	24,494	
2006	52	40,824	0	
2007	51	16,329	24,494	
2008	35	8,165	32,659	
2009	42	8,165	32,659	
2010	38	8,165	24,494	General category IFQs established ACL management implemented
2011	32	16,329	16,329	
2012	34	20,412	12,247	
2013	33	7,502	3,787	
2014	31	5,210	5,210	
2015	30.86	0	23,133	
2016	34.55	0	23,133	
2017	30.41	16,329	16,329	
2018	24	32,659	16,329	
2019	24	32,659	24,494	
2020	24	24,494	16,329	
2021	24	24,494	8,165	
2022	24	20,412	0	
2023	20	10,886	5,443	

Table 2.2. (a) Sea Scallop landings (mt meats) in the Mid-Atlantic (MA), Georges Bank (GB), and Southern New England (SNE) regions for 1964-2023. Landings from area unknown were prorated to region.

	MA				GB				SNE			
	Dredge	Trawls	Other	Sum	Dredge	Trawls	Other	Sum	Dredge	Trawls	Other	Sum
1964	140	0	0	140	6241	0	0	6241	0	52	0	52
1965	3997	2	0	3999	1480	3	0	1483	1	0	0	1
1966	4064	0	0	4064	883	0	0	884	5	0	0	5
1967	1877	0	0	1877	1217	4	0	1221	3	0	0	3
1968	2450	0	0	2450	993	0	0	994	42	0	0	42
1969	858	6	0	864	1316	8	0	1324	6	0	0	6
1970	460	14	0	474	1410	5	0	1415	5	0	0	5
1971	277	0	0	278	1311	18	0	1329	3	0	0	3
1972	655	5	0	660	816	5	0	821	0	0	0	0
1973	248	4	0	251	1065	15	0	1080	0	0	0	1
1974	939	0	0	939	911	15	0	926	3	0	0	3
1975	1538	52	0	1590	844	13	0	857	10	8	0	18
1976	2973	319	0	3292	1723	38	0	1761	2	1	0	3
1977	2570	27	0	2597	4709	27	0	4735	5	0	0	5
1978	4177	21	0	4199	5532	37	0	5569	23	1	0	24
1979	2895	34	1	2931	6257	25	3	6284	23	0	0	23
1980	2023	12	0	2034	5380	34	2	5416	74	0	0	74
1981	744	5	0	748	7786	56	0	7843	51	1	0	51
1982	1639	6	2	1647	6203	119	0	6322	89	0	0	89
1983	3181	19	8	3207	4247	32	0	4280	148	0	0	149
1984	3683	27	2	3711	3014	29	0	3043	125	3	0	128
1985	3273	51	1	3325	2860	34	0	2894	32	0	0	32
1986	3308	101	0	3409	4428	10	0	4438	25	2	0	27
1987	7538	315	1	7854	4821	30	0	4850	16	1	0	18
1988	5815	402	2	6219	6036	18	0	6054	23	4	0	27
1989	7589	430	2	8021	5727	25	0	5752	167	5	0	172
1990	5999	478	4	6480	10059	10	0	10069	107	5	0	112
1991	6234	809	9	7052	9355	77	0	9432	61	14	0	75
1992	4432	564	5	5002	8333	7	0	8340	153	4	0	157
1993	2432	393	3	2828	3652	18	0	3669	50	1	0	51
1994	5132	747	75	5955	1182	6	1	1189	28	0	0	28
1995	5655	783	162	6600	984	2	16	1002	28	1	0	29
1996	4475	653	4	5133	2130	7	4	2141	45	5	0	50
1997	2737	382	0	3119	2349	7	0	2356	44	10	3	57
1998	2452	569	6	3027	2046	15	0	2062	58	3	0	61
1999	3647	961	1	4609	5172	5	0	5177	58	1	0	59
2000	8131	1214	0	9345	4907	37	6	4950	80	2	2	84
2001	14103	1567	9	15678	4863	33	28	4924	26	38	1	65
2002	15940	1428	93	17462	5957	29	7	5992	8	12	0	19
2003	19001	1231	0	20231	4853	19	0	4872	39	2	2	43
2004	22303	1196	66	23565	4233	139	34	4406	730	140	42	912
2005	13804	1045	989	15837	8561	49	79	8689	681	9	143	833
2006	8171	772	342	9285	15487	42	46	15575	1624	7	21	1651
2007	16065	392	290	16747	9369	45	57	9471	163	5	16	184
2008	16583	555	170	17308	6341	22	81	6445	277	2	50	329
2009	18134	602	224	18960	6776	21	60	6857	263	1	6	270
2010	19246	281	96	19623	5819	18	48	5885	177	7	25	209
2011	17385	321	200	17906	8119	15	137	8270	176	24	21	220
2012	11212	277	167	11656	13685	37	16	13739	63	2	30	95
2013	6191	228	45	6464	11566	27	21	11614	68	11	2	81
2014	9321	220	40	9582	5271	12	4	5287	86	5	0	91
2015	10447	172	31	10649	5102	5	0	5108	74	5	1	79
2016	13226	331	142	13699	3846	3	151	4001	99	2	1	103
2017	15154	179	84	15417	6877	12	77	6966	115	1	0	116
2018	9842	119	178	10139	15682	2	122	15806	102	2	3	107
2019	10452	115	117	10684	16279	11	76	16367	147	1	12	159
2020	12852	70	92	13013	8545	0	33	8578	161	0	0	162
2021	4537	40	18	4595	14096	0	61	14157	169	0	1	170
2022	2051	27	13	2091	10869	26	0	10895	375	1	0	377
2023	651	12	6	669	10736	0	4	10740	373	0	0	374

Table 2.2. (b) Sea Scallop landings (mt meats) in the Gulf of Maine (GOM) and total across all regions for 1964-2023. Landings from area unknown were prorated to region.

	GOM				TOTAL			
	Dredge	Trawls	Other	Sum	Dredge	Trawls	Other	Sum
1964	208	0	0	208	6589	52	0	6642
1965	115	0	0	115	5593	5	0	5598
1966	102	0	0	102	5055	1	0	5056
1967	80	0	0	80	3177	4	0	3182
1968	113	0	0	113	3599	0	0	3599
1969	122	1	0	123	2302	14	0	2317
1970	132	0	0	132	2006	19	0	2026
1971	358	4	0	362	1949	22	0	1971
1972	524	1	0	525	1995	11	0	2006
1973	460	0	0	460	1772	19	1	1792
1974	223	0	0	223	2075	16	0	2091
1975	741	6	0	746	3132	80	0	3212
1976	363	3	0	366	5061	361	0	5422
1977	254	4	0	258	7536	58	0	7595
1978	242	1	0	243	9974	60	0	10035
1979	405	5	1	411	9580	64	5	9649
1980	1491	122	3	1616	8967	169	4	9141
1981	1229	73	7	1309	9810	135	7	9952
1982	631	28	5	664	8562	153	7	8722
1983	823	72	7	902	8399	124	15	8538
1984	651	18	10	678	7473	76	11	7560
1985	428	3	10	441	6593	88	11	6692
1986	308	2	6	316	8068	115	7	8190
1987	374	0	8	382	12749	346	9	13104
1988	506	7	13	526	12380	430	16	12826
1989	624	0	44	668	14107	460	45	14612
1990	556	0	28	584	16720	493	32	17246
1991	536	3	75	614	16187	903	84	17173
1992	698	2	45	745	13616	578	50	14245
1993	779	2	32	813	6913	413	36	7361
1994	410	6	10	426	6752	759	86	7598
1995	343	7	13	362	7009	793	191	7994
1996	545	5	13	563	7195	671	21	7887
1997	673	5	23	701	5802	405	26	6233
1998	393	4	18	415	4949	591	24	5565
1999	267	2	5	275	9144	969	7	10119
2000	161	21	61	243	13280	1274	68	14623
2001	334	6	13	352	19325	1644	51	21020
2002	386	18	2	405	22291	1486	102	23879
2003	197	3	53	253	24091	1254	56	25400
2004	164	12	1	177	27430	1488	144	29062
2005	155	11	82	248	23200	1115	1293	25608
2006	131	3	16	150	25412	825	424	26661
2007	93	8	25	125	25689	449	388	26527
2008	92	12	16	120	23294	592	317	24203
2009	59	6	6	71	25232	630	295	26158
2010	151	14	4	168	25393	319	172	25885
2011	225	17	3	244	25905	376	360	26641
2012	373	22	1	397	25333	339	215	25886
2013	441	43	13	498	18266	310	82	18657
2014	283	88	11	382	14962	324	56	15343
2015	264	98	9	371	15887	279	40	16206
2016	553	78	5	636	17724	415	299	18438
2017	800	148	10	957	22946	340	171	23457
2018	261	72	50	383	25888	194	353	26435
2019	377	24	38	439	27254	151	243	27648
2020	401	62	13	476	21959	132	138	22229
2021	691	67	15	772	19492	108	95	19695
2022	1029	48	19	1096	14324	101	33	14459
2023	897	16	13	926	12658	29	23	12710

Table 2.3. Estimated discards (Disc, mt meats), coefficient of variation (CV), and percent of landings (% L) by region for the Mid-Atlantic (MA), Southern New England (SNE), Georges Bank (GB), the Gulf of Maine (GOM), and total. 1989-2009 values are from SARC 59 (NEFSC 2014); 2010-2017 values are from SARC 65 (NEFSC 2018); 2018-2023 values are from CAMS. Block Island (area 329) discards are included with SNE. Discards for SNE and GOM were not computed for 1989-2009 during SARC 59 because they were considered minimal.

Year	MA			SNE			GB			GOM			Total		
	Disc	CV	% L	Disc	CV	% L	Disc	CV	% L	Disc	CV	% L	Disc	CV	% L
1989	213		2.7				4		0.1				217		1.5
1990	8		0.1				1		0.0				9		0.1
1991	11		0.2				5		0.1				16		0.1
1992	131		2.6				465		5.6				596		4.2
1993	22		0.8				346		9.4				368		5.0
1994	703		11.8				10		0.8				713		9.4
1995	495		7.5				23		2.3				518		6.5
1996	38		0.7				116		5.4				154		2.0
1997	11		0.4				46		2.0				57		0.9
1998	53		1.8				4		0.2				57		1.0
1999	114		2.5				142		2.7				256		2.5
2000	813		8.7				991		20.0				1,804		12.3
2001	1,969		12.6				531		10.8				2,500		11.9
2002	1,970		11.3				107		1.8				2,077		8.7
2003	2,244		11.1				332		6.8				2,576		10.1
2004	2,559		10.9				102		2.3				2,661		9.2
2005	424		2.7				238		2.7				662		2.6
2006	244		2.6				378		2.4				622		2.3
2007	294		1.8				236		2.5				530		2.0
2008	457		2.6				341		5.3				798		3.3
2009	1,013		5.3				398		5.8				1,402		5.4
2010	730	0.18	3.7	2	1.18	1.0	715	0.17	12.1	3	0.67	1.8	1,450	0.12	5.6
2011	537	0.14	3.0	19	0.44	8.6	536	0.21	6.5	2	0	0.8	1,094	0.13	4.1
2012	278	0.10	2.4	0	-	0.0	890	0.10	6.5	0	-	0.0	1,168	0.08	4.5
2013	157	0.17	2.4	5	0.33	6.2	357	0.14	3.1	6	0.31	1.2	525	0.11	2.8
2014	70	0.16	0.7	20	0.47	21.9	220	0.16	4.2	9	0.40	2.4	319	0.12	2.1
2015	562	0.11	5.3	11	0.45	13.9	191	0.18	3.7	2	0.21	0.5	766	0.09	4.7
2016	2,074	0.07	15.1	18	0.24	17.6	98	0.20	2.4	6	0.30	0.9	2,190	0.07	11.9
2017	931	0.22	6.0	3	0.23	2.6	479	0.11	6.9	34	0.22	3.6	1,447	0.15	6.2
2018	429	0.15	4.2	22	0.34	20.6	852	0.15	5.4	10	0.28	2.6	1,313	0.11	5.0
2019	461	0.12	4.3	37	0.34	23.3	1,235	0.12	7.5	2	0.30	0.5	1,735	0.09	6.3
2020	678	0.19	5.2	44	0.77	27.2	1,059	0.30	12.3	2	0.73	0.4	1,782	0.19	8.0
2021	272	0.21	5.9	25	0.52	14.7	1,427	0.13	10.1	17	0.34	2.2	1,741	0.11	8.8
2022	106	0.24	5.1	26	0.47	6.9	519	0.15	4.8	54	0.18	4.9	704	0.12	4.9
2023	78	0.92	11.7	59	1.12	15.8	728	0.22	6.8	51	0.17	5.5	916	0.20	7.2

Figures

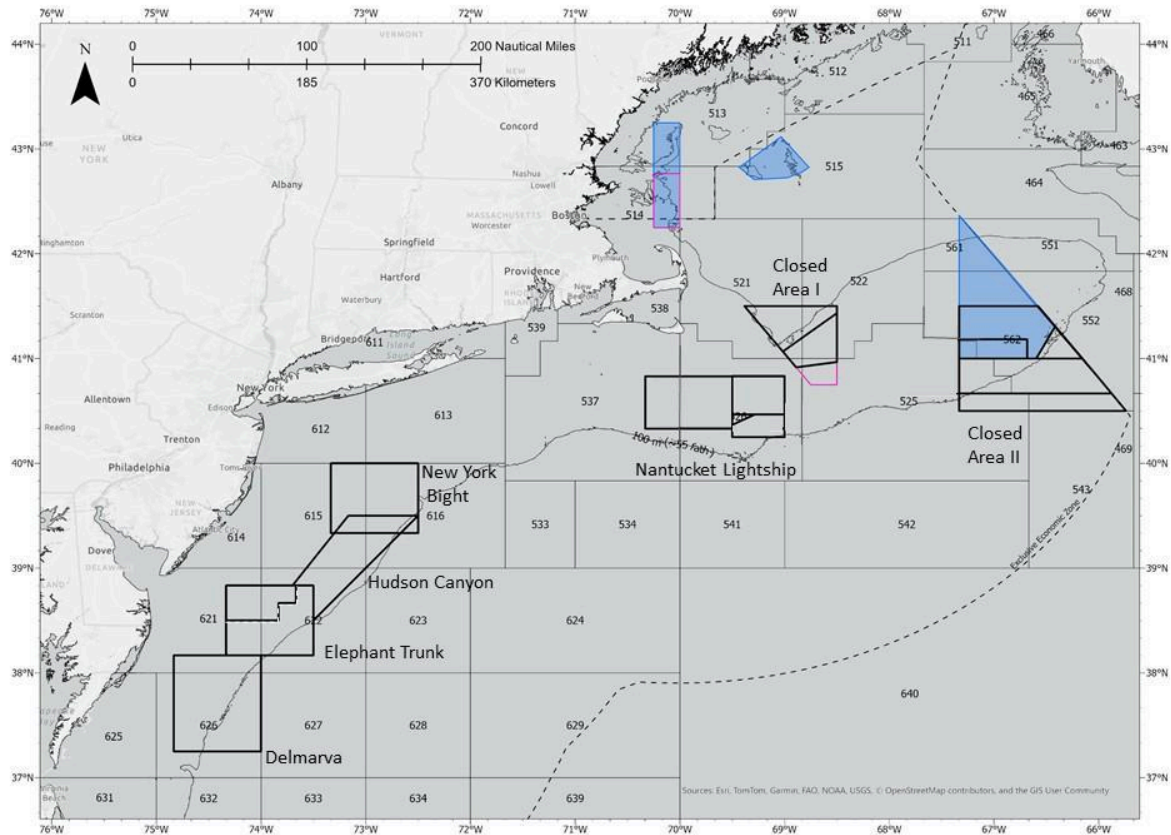


Figure 2.1. Map of scallop rotational access areas (thick black outlines). For additional context, fishing statistical areas (thin black lines), groundfish closure areas (blue polygons), dedicated habitat research areas (pink outlines) and the US Exclusive Economic Zone (dotted line) are also included.

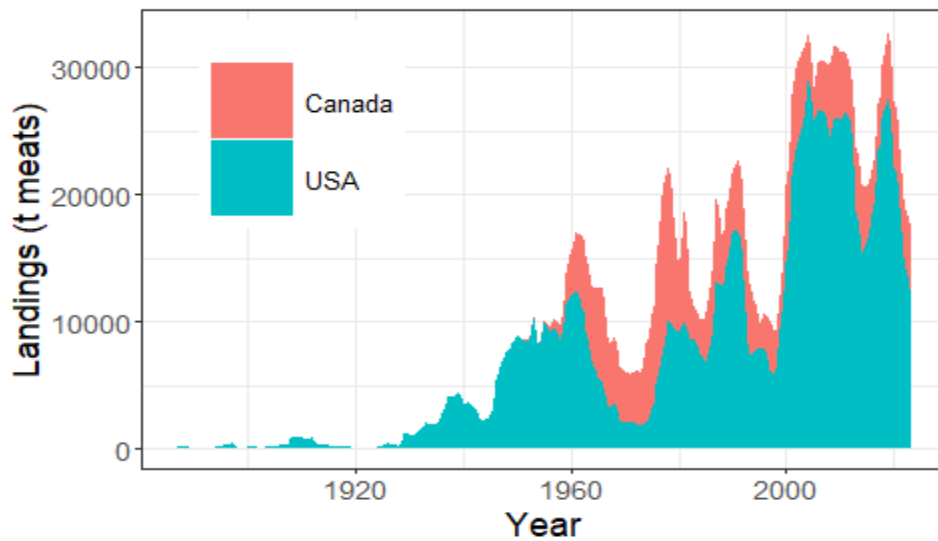


Figure 2.2. Long-term sea scallop landings in NAFO areas 5 and 6, comprising all U.S. scallop grounds as well as the Canadian portion of Georges Bank. The split between U.S. and Canadian landings prior to 1985 is with respect to port landed, rather than fishing location.

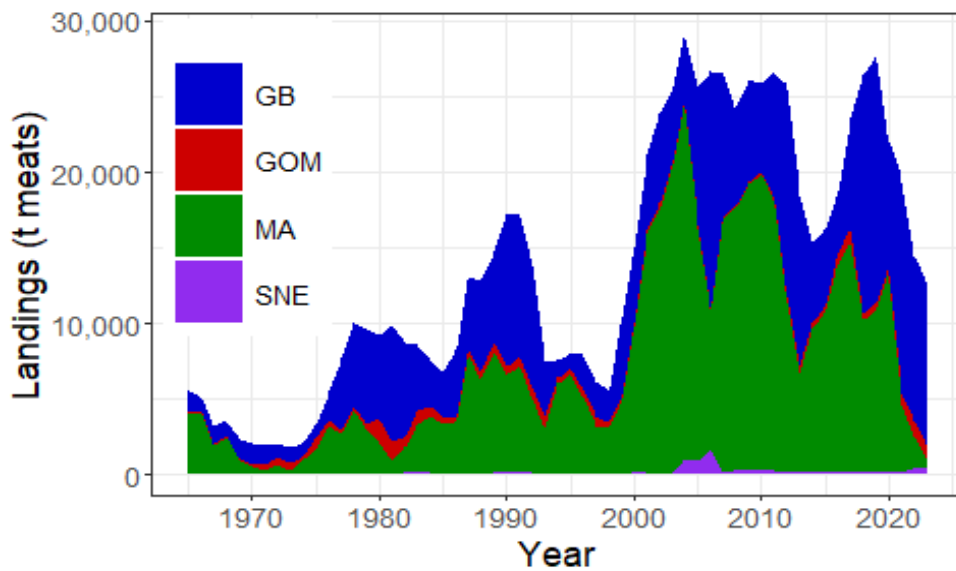


Figure 2.3. U.S. sea scallop landings (mt meats) for the Mid-Atlantic (MA), Georges Bank (GB), Southern New England (SNE), and Gulf of Maine (GOM) regions for 1965-2023. Landings where the area was unknown were prorated

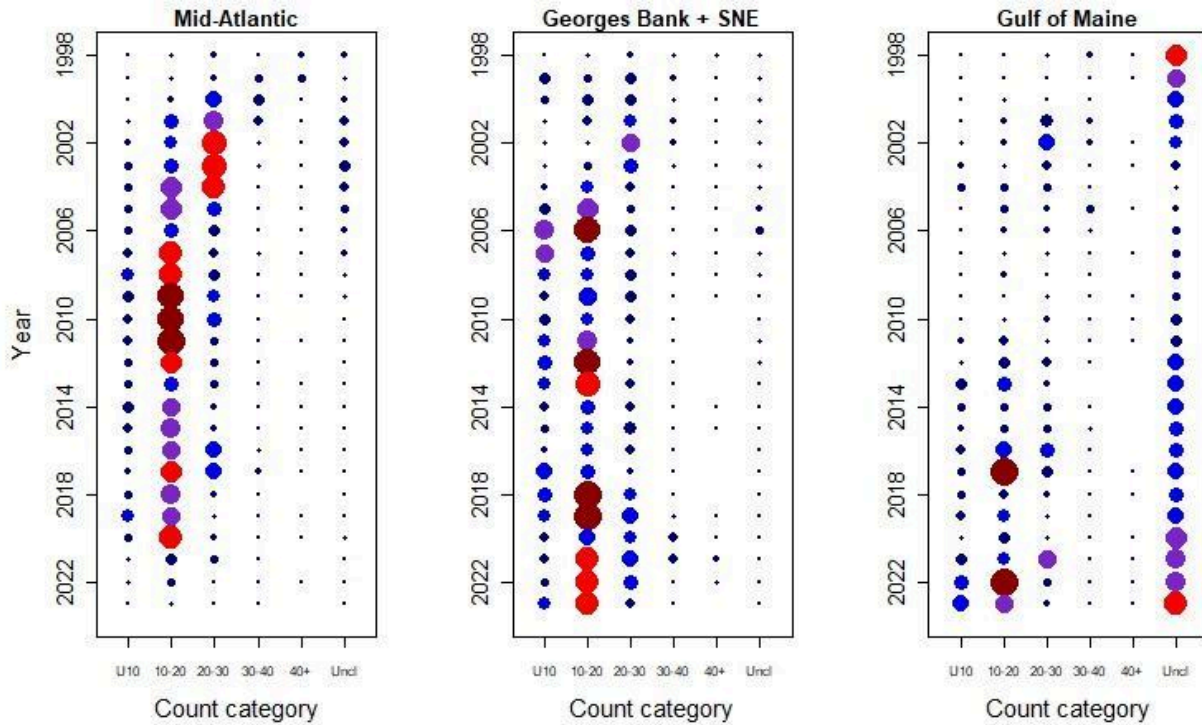


Figure 2.4. U.S. sea scallop proportion of landings by region and meat count for 1998-2023. Market categories are U10 (>45 g), 10-20 (23 to 45 g), 20-30 (15-23 g), 30-40 (11 to 15 g), 40+ (less than 11 g), and unclassified. Southern New England is abbreviated as ‘SNE’.

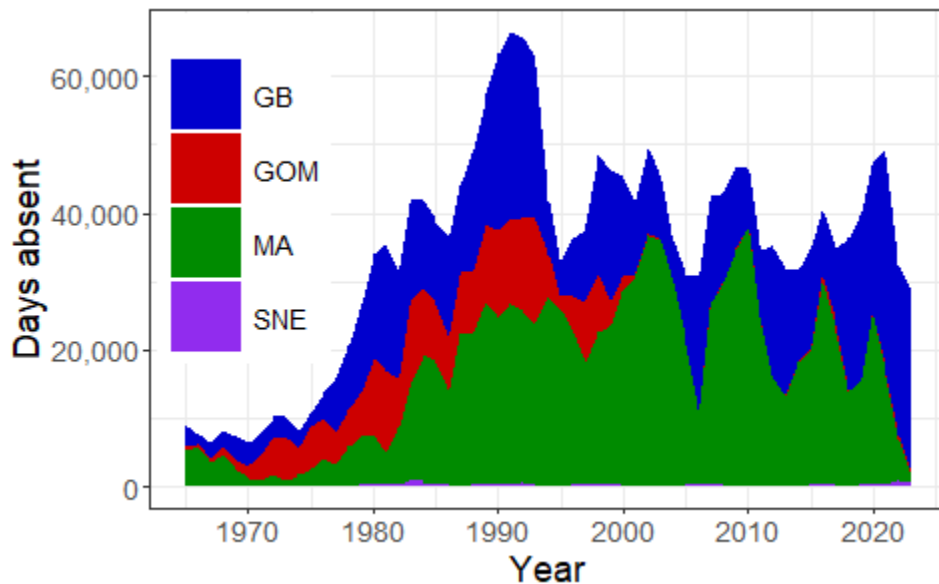


Figure 2.5. Limited access sea scallop effort, in days absent for the Mid-Atlantic (MA), Georges Bank (GB), Southern New England (SNE), and Gulf of Maine (GOM) regions for 1965-2023. Much of the effort in the Gulf of Maine prior to 2000 is due to vessels primarily targeting sea urchins with dredge gear, but also landed sea scallops.

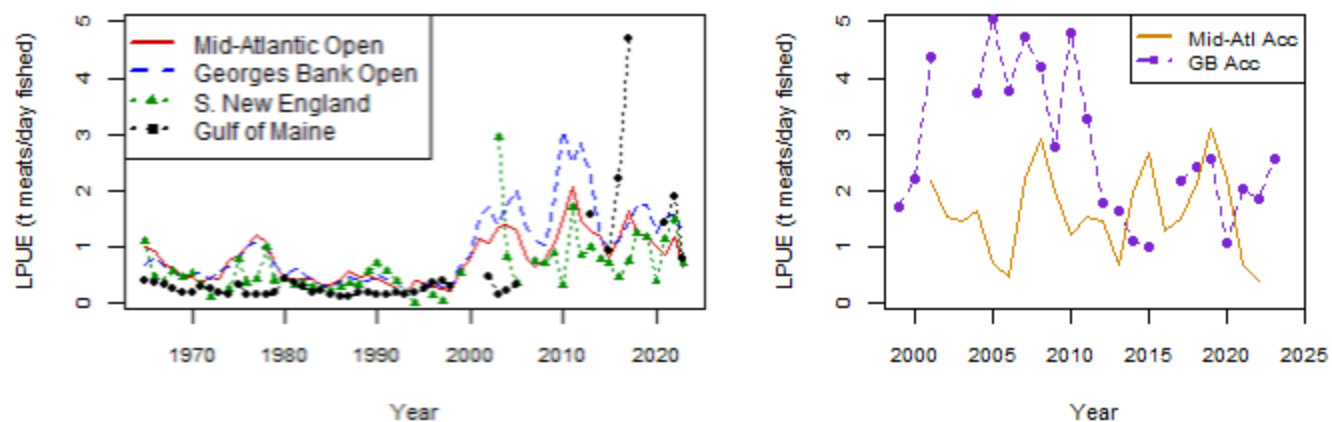


Figure 2.6. LPUE of limited access sea scallop vessels (mt per day fished), by region, not including access areas (i.e., open) after 1998 for the Mid-Atlantic, Georges Bank, Southern New England, and Gulf of Maine regions (1965-2023, left) and in access areas ('Acc') for the Mid-Atlantic, Georges Bank (1998-2023, right).

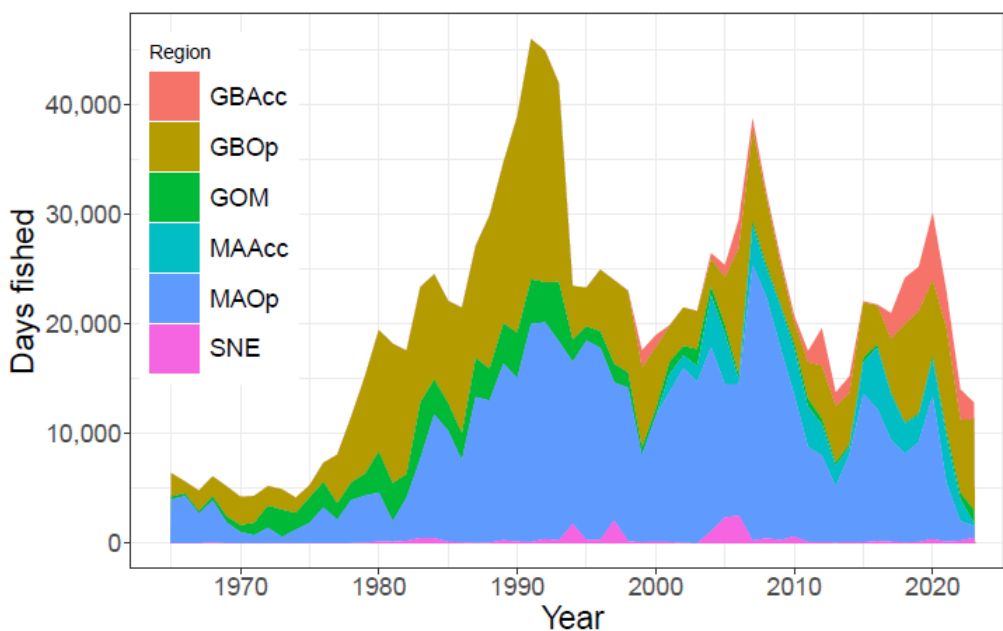


Figure 2.7. Sea scallop fishing effort, in days fished (equivalent) for limited access vessels for 1965-2023. All areas are considered open prior to 1995.

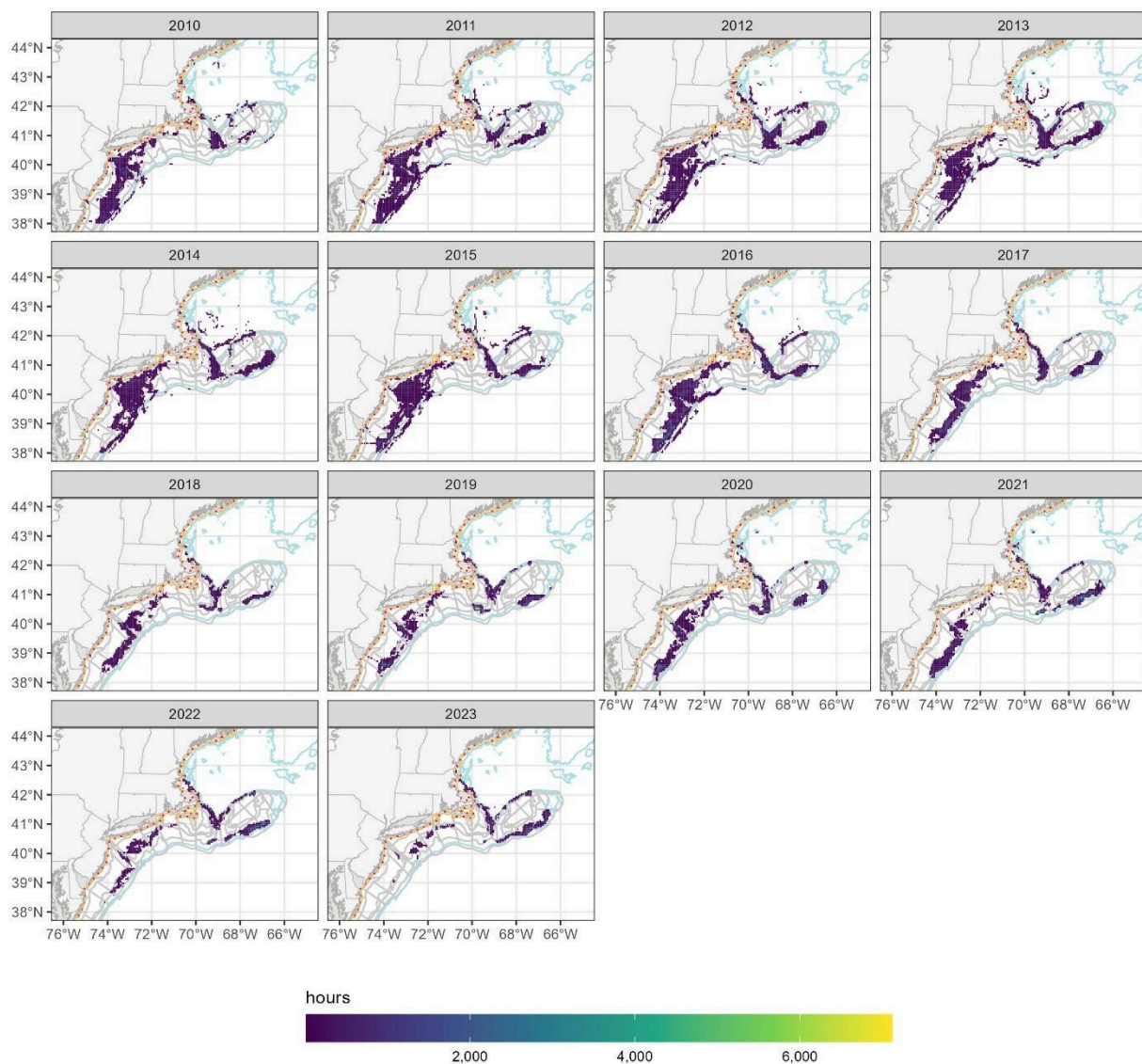


Figure 2.8. Vessel Monitoring System (VMS) plots of scallop fishing effort for 2010-2023 for all fleets combined. Data are summarized by fishing year, with fishing defined as a 2-5.5 knots speed filter, binned at 3 minutes cell size, with a minimum of 20 hours per cell.

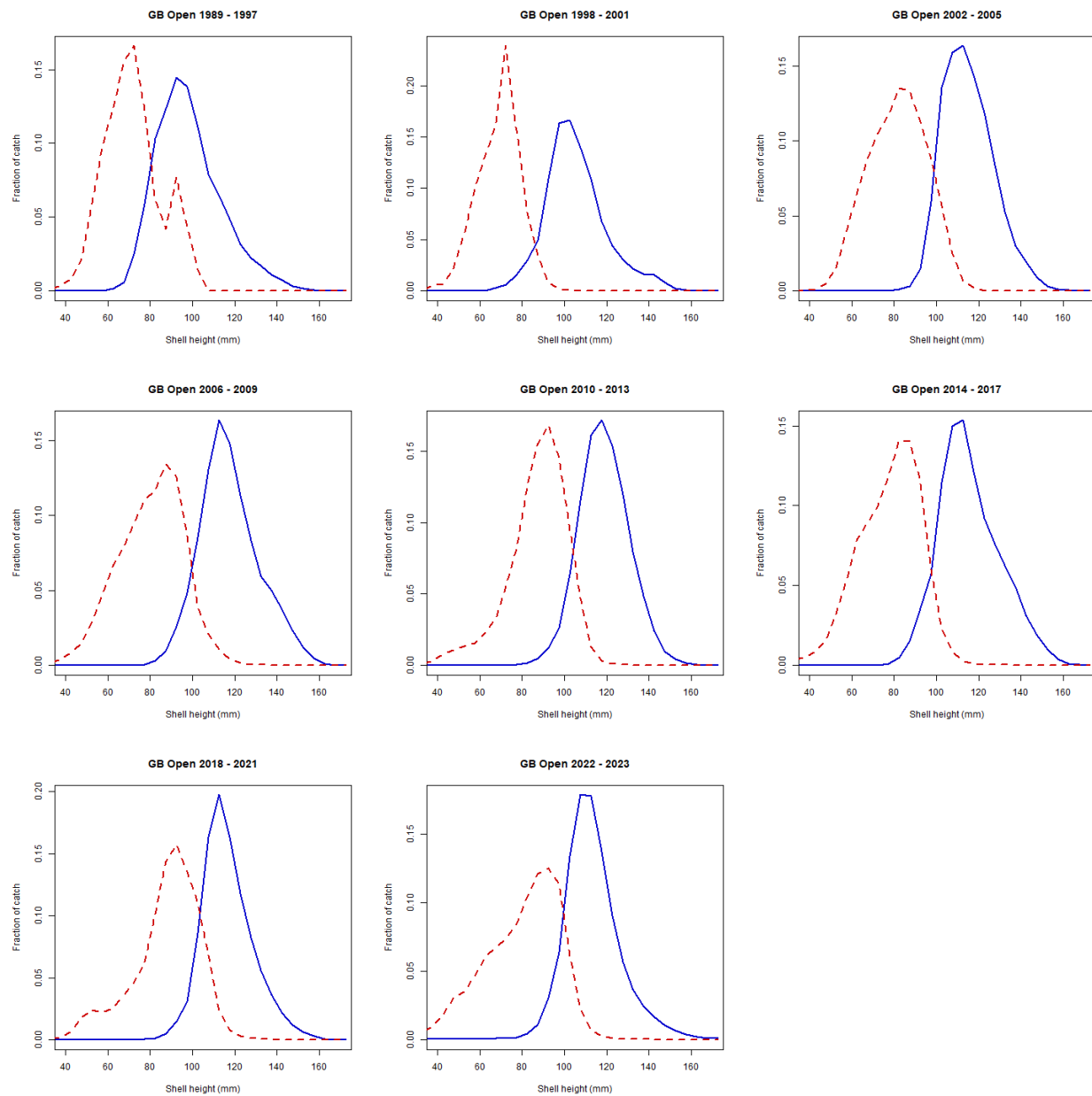


Figure 2.9 (a). Size frequencies of commercial catch for Georges Bank open areas during the specified periods. The solid blue line is retained scallops, and the dotted line is discarded scallops.

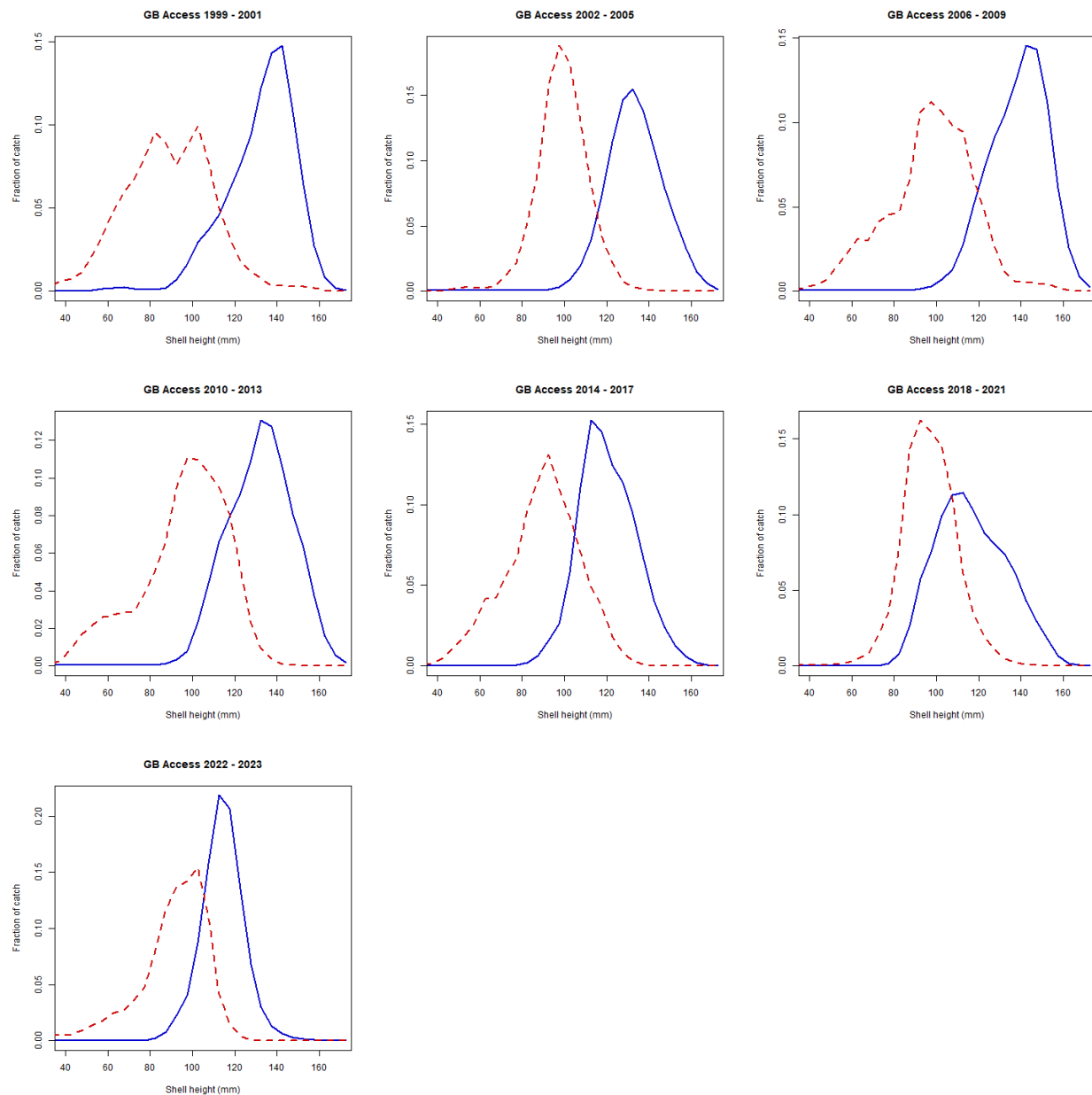


Figure 2.9 (b). Size frequencies of commercial catch for Georges Bank access areas during the specified periods. The solid blue line is retained scallops, and the dotted line is discarded scallops.

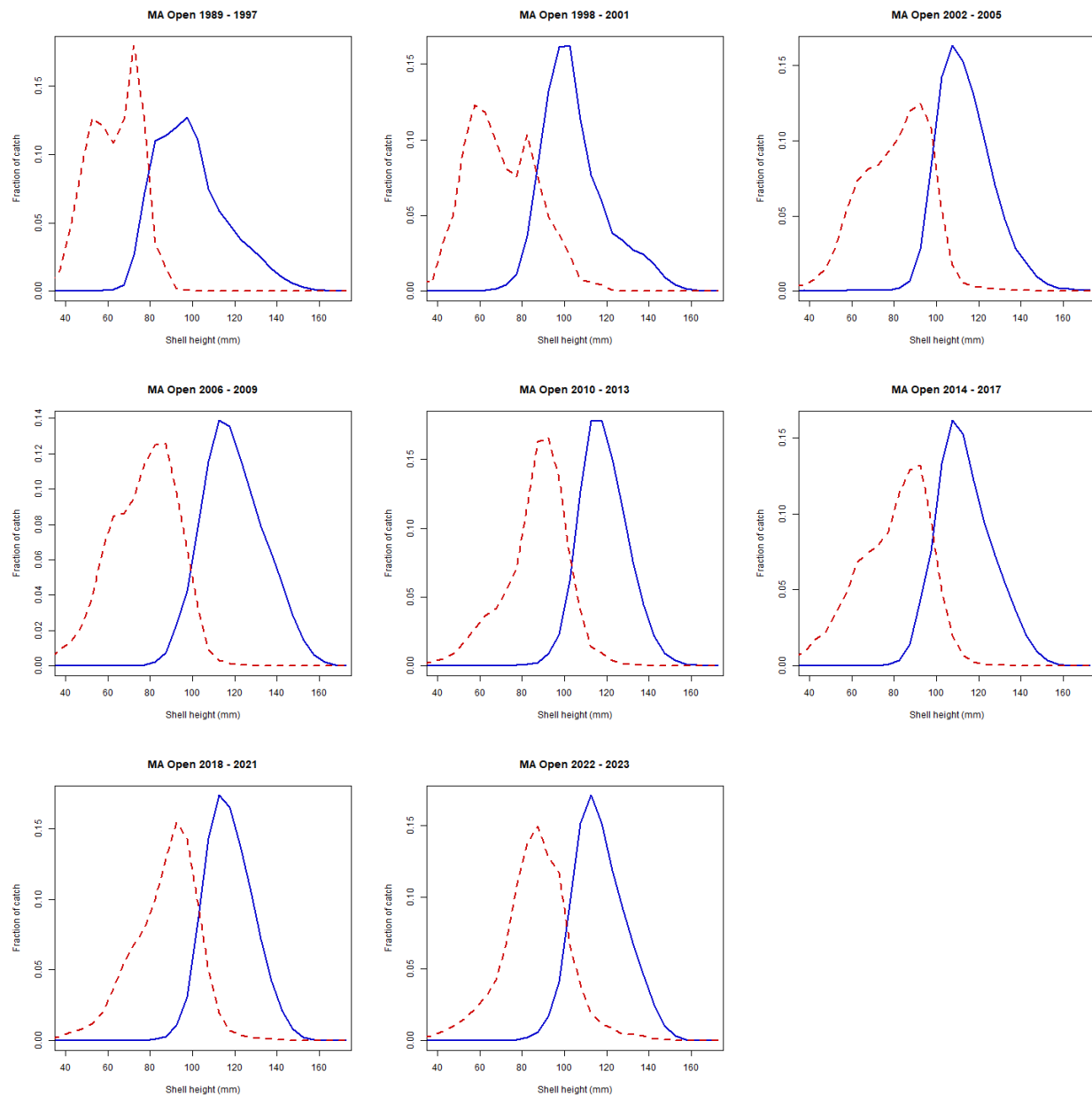


Figure 2.9 (c). Size frequencies of commercial catch for Mid-Atlantic open areas during the specified periods. The solid blue line is retained scallops, and the dotted line is discarded scallops.

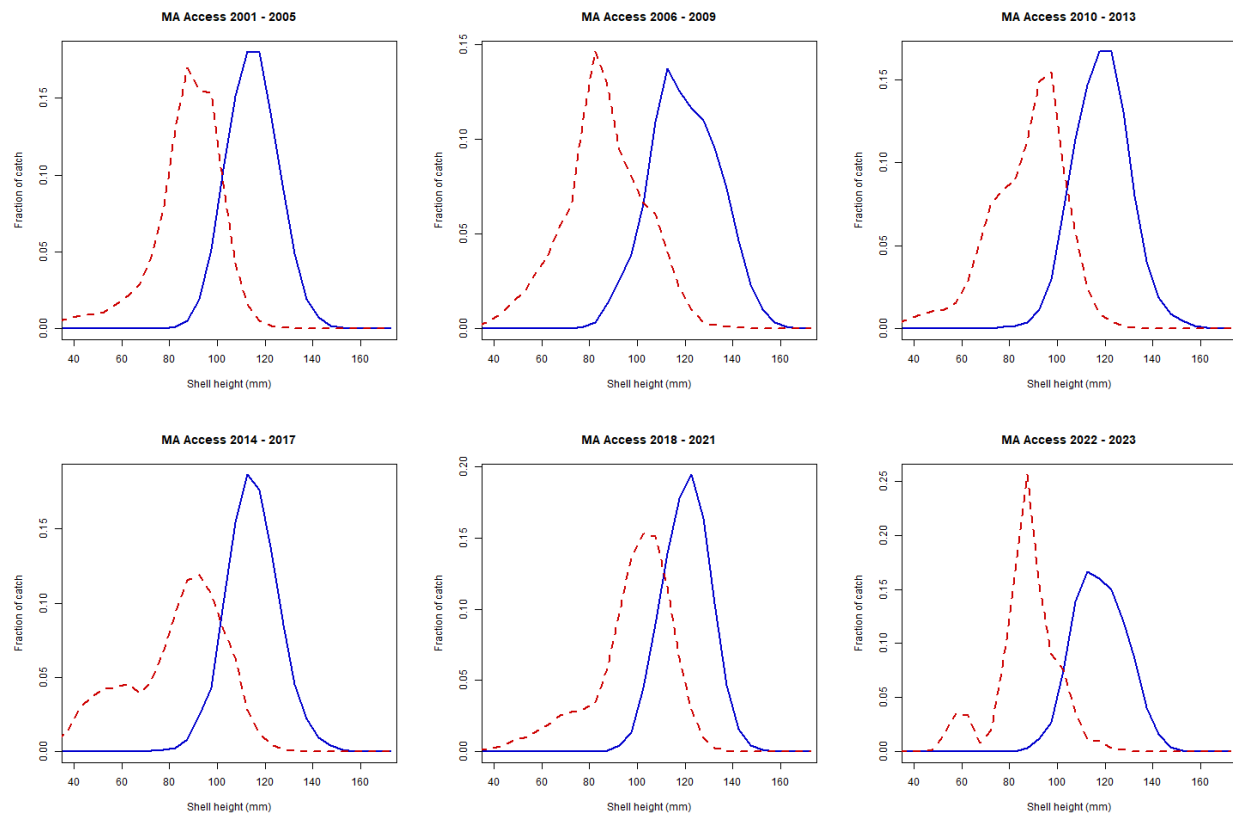


Figure 2.9 (d). Size frequencies of commercial catch for Mid-Atlantic access areas during the specified periods. The solid blue line is retained scallops, and the dotted line is discarded scallops.

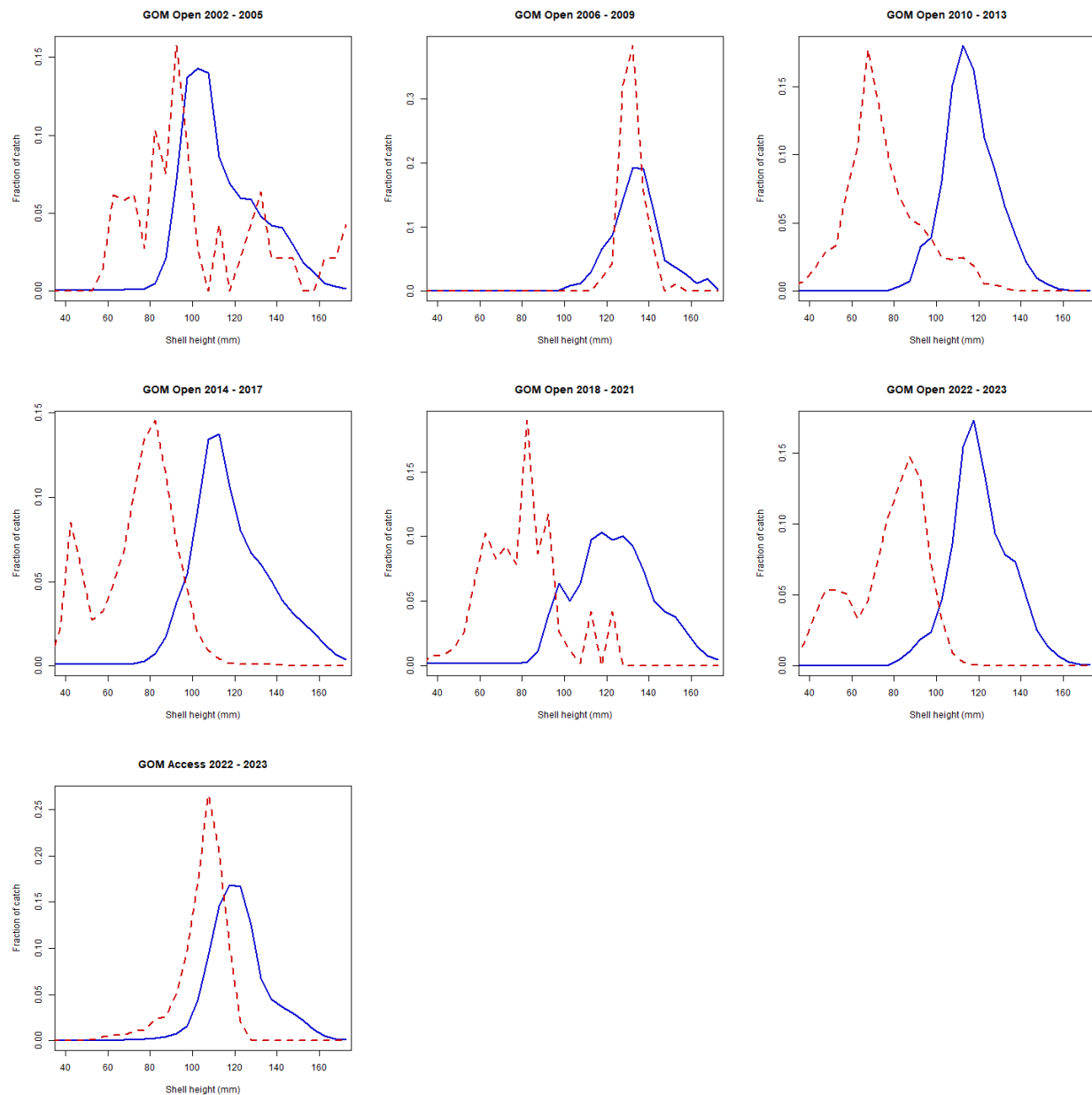


Figure 2.9 (e). Size frequencies of commercial catch for Gulf of Maine open areas (top two rows) and the Northern Gulf of Maine area during the specified periods. The solid blue line is retained scallops, and the dotted line is discarded scallops.